

NEW CASTLE, DE, USA

WMO: 724180

Lat: 39.674N

Lon: 75.606W

Elev: 24

StdP: 101.04

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13781

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.6	-8.4	-19.7	0.7	-9.0	-17.3	0.8	-6.9	12.8	0.0	11.9	0.0	4.8	290	0.498

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.5	33.4	24.0	31.9	23.4	30.6	22.9	25.7	30.7	24.9	29.6	24.3	28.4	4.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.2	19.2	27.7	23.6	18.4	27.1	22.9	17.7	26.5	78.8	30.9	75.7	29.6	73.1	28.4	29.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
11.1	9.4	8.4	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-13.6	35.6	2.7	1.6	-15.6	36.8	-17.1	37.7	-18.6	38.6	-20.6	39.8		
			-14.6	27.1	2.6	0.9	-16.5	27.8	-18.0	28.3	-19.5	28.9	-21.4	29.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.0	0.6	1.8	6.1	12.2	17.4	22.5	25.1	24.2	20.5	14.0	8.0	3.4
	DBStd	9.57	5.51	5.06	5.01	4.53	4.15	3.41	2.72	2.68	3.60	4.41	4.43	4.76
	HDD10.0	1008	296	232	142	26	1	0	0	0	0	12	89	211
	HDD18.3	2598	551	462	379	191	71	7	0	1	19	147	309	462
	CDD10.0	2119	3	4	21	91	228	374	468	439	316	137	30	7
	CDD18.3	668	0	0	1	7	40	131	209	181	84	14	0	0
	CDH23.3	5668	0	0	9	79	331	1094	1977	1524	576	77	1	0
	CDH26.7	1894	0	0	1	19	92	354	764	499	152	13	0	0
Wind	WSAvg	3.8	4.2	4.3	4.6	4.4	3.7	3.5	3.2	3.1	3.4	3.5	3.8	3.9
Precipitation	PrecAvg	1050	76	71	97	84	98	94	103	87	102	71	77	86
	PrecMax	1405	214	178	233	173	188	252	321	284	322	163	199	201
	PrecMin	632	13	11	21	9	6	5	23	6	21	2	12	20
	PrecStd	185	40	37	47	40	45	49	58	53	66	38	48	49
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.5	19.6	24.2	29.1	31.7	34.1	35.8	34.5	32.7	28.2	21.9	19.4
		MCWB	14.2	14.2	16.0	18.7	22.3	24.1	25.2	24.5	22.8	21.1	16.8	16.3
	2%	DB	13.9	15.2	19.9	25.7	29.2	32.0	33.8	32.4	30.4	25.5	19.4	15.8
		MCWB	11.5	10.8	13.4	17.4	21.3	23.1	24.3	24.0	22.2	19.6	15.2	13.4
	5%	DB	11.1	11.7	16.8	22.7	27.1	30.2	32.2	30.9	28.2	23.2	17.3	13.0
		MCWB	8.8	7.7	11.3	15.8	20.1	22.5	23.8	23.2	21.4	18.4	13.8	10.8
	10%	DB	8.3	9.2	13.8	20.1	24.8	28.7	30.8	29.4	26.6	21.3	15.2	10.6
		MCWB	6.0	6.0	9.6	14.2	18.8	21.7	23.3	22.4	20.6	17.3	12.4	8.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.0	15.7	17.5	20.7	23.8	25.5	27.3	26.4	25.2	23.0	18.4	17.0
		MCDB	16.6	19.0	23.1	26.0	29.6	31.4	32.2	31.7	29.3	25.6	20.7	18.6
	2%	WB	12.2	11.6	14.7	18.6	22.5	24.6	25.9	25.4	24.2	21.3	16.6	13.9
		MCDB	13.1	13.9	18.3	23.7	27.5	29.6	31.2	30.0	27.3	24.1	18.2	15.5
	5%	WB	9.0	8.5	12.4	16.9	21.2	23.9	25.2	24.7	23.3	19.8	14.7	11.4
		MCDB	10.6	10.9	15.6	21.2	25.6	28.4	30.2	28.6	26.2	22.2	16.5	12.6
	10%	WB	6.4	6.3	10.2	15.3	20.0	23.1	24.5	24.0	22.3	18.0	13.0	8.8
		MCDB	8.1	8.9	13.3	19.2	23.7	27.0	28.9	27.5	25.1	20.4	14.7	10.2
Mean Daily Temperature Range	5% DB	MDBR	7.9	8.7	9.7	10.7	10.2	9.7	9.5	9.3	9.9	10.0	9.4	7.9
		MCDBR	11.4	12.7	14.1	14.3	12.6	11.2	11.0	10.4	11.2	11.7	11.8	10.9
		MCWBR	9.5	9.3	9.0	7.7	6.1	5.0	4.2	4.4	5.0	6.5	8.2	9.4
	5% WB	MCDBR	10.8	11.7	12.8	12.5	11.0	9.8	9.5	8.8	8.6	9.6	9.8	9.9
		MCWBR	9.9	9.7	9.5	7.8	6.0	4.9	4.1	4.2	4.7	6.5	8.0	9.6
Clear-Sky Solar Irradiance	taub	0.315	0.321	0.363	0.403	0.447	0.511	0.519	0.514	0.425	0.363	0.327	0.311	
	taud	2.445	2.417	2.361	2.276	2.186	2.048	2.039	2.039	2.264	2.443	2.503	2.503	
	Ebn at Noon	851	898	890	873	838	782	770	762	815	839	832	831	
	Edh at Noon	81	95	112	130	145	167	167	163	122	92	76	71	
All-Sky Solar Radiation	RadAvg	1.96	2.85	3.75	4.86	5.44	5.96	6.01	5.28	4.35	3.15	2.25	1.71	
	RadStd	0.17	0.32	0.30	0.40	0.67	0.45	0.27	0.32	0.48	0.33	0.20	0.17	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+0.50	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+115	+69
Regional (79 neighbors)		+0.46	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+133	+78

Nomenclature: See separate page